

Abstract:

Scientists at the Lenski Lab discovered a mutation in *Escherichia coli* populations, cultivated in a glucose-limited medium, that enabled them to metabolize citrate. This study analyzes additional data on citrate concentrations in four *E. coli* populations (A, B, C, D), representing different generations of the citrate-metabolizing mutation. Citrate concentrations were measured as changes in Optical Density at 420 nm (OD420) over a 24-hour period, with samples taken at 0, 1, 2, 3, 4, 6, 9, and 24 hours. Using R, I calculated the mean and standard deviation of OD420 for five replicates of each population. Results showed that populations A and B (30,000 and 32,000 generations) had relatively stable OD420 values, indicating minimal citrate utilization. In contrast, populations C and D (34,000 and 40,000 generations) exhibited a significant drop in OD420, reflecting increased citrate consumption. These findings suggest that the citrate-metabolizing mutation became more effective in later generations, allowing populations C and D to exploit citrate more efficiently than earlier generations.

Abstract, First Draft:

Scientists at the Lenski Lab, cultivating *E. coli* in a reduced glucose medium that also contained citrate, a mutation allowed one population of *E. coli* to metabolize citrate. This study investigates the evolution of that trait by analyzing an additional dataset generated by the Lenski Lab. The dataset shows changes in citrate concentrations in the culture medium of four *E. coli* populations (A, B, C, and D) over a period of 24 hours and measured the changes in citrate concentrations as changed in Optical Density at 420 nm (OD420). The populations (A, B, C, and D) are taken from different generations of the *E. coli* population that initially developed the mutation to metabolize citrate. In this study, I analyze the dataset using the R programming language to calculate the mean and standard deviation for each generation in each replicant. I found a low variance in the OD420 values of generation A and B, which suggests that the citrate concentrations stayed fairly consistent over the 24 hour period. In generation C and D I found a lower mean OD420 value and a high variance, which suggests that the citrate concentration decreased during the 24 hour period. Plotting the OD420 values vs time showed a sudden drop in citrate concentration for generations C and D during the 24 hour period, suggesting that these generations became more efficient at metabolizing citrate. These results were consistent across all 5 replicants.

How I used ChatGPT:

I gave ChatGPT the relevant context (the research paper and assignment) along with my first draft and asked it to revise the draft, offer feedback and suggestions. ChatGPT gave me suggestions that made the Abstract more clear and concise, helped me better understand the paper, and interpret my findings. I also asked ChatGPT questions about R programming, while it mostly gave me code snippets that caused errors, it did help me write my plots to .png files.

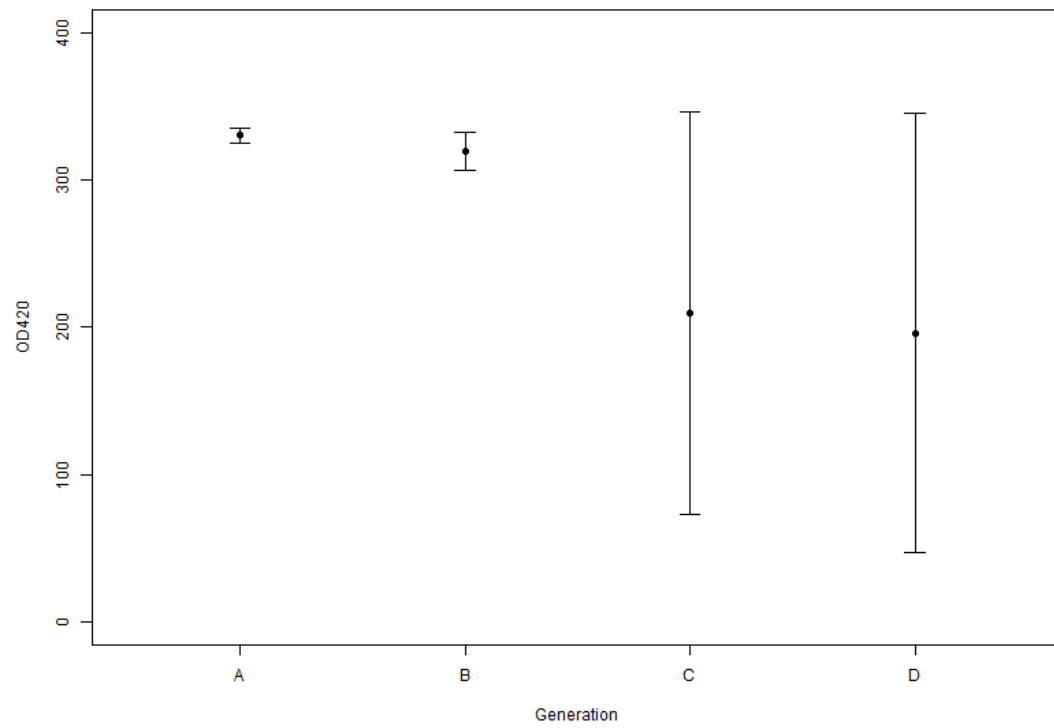
TABLE-2: Mean and Standard Deviation for each generation across all replicants combined.

	Generation	mean	stddev
1	A	319.9	16.3
2	B	306.1	13.8
3	C	203.4	122.7
4	D	192.5	134.9

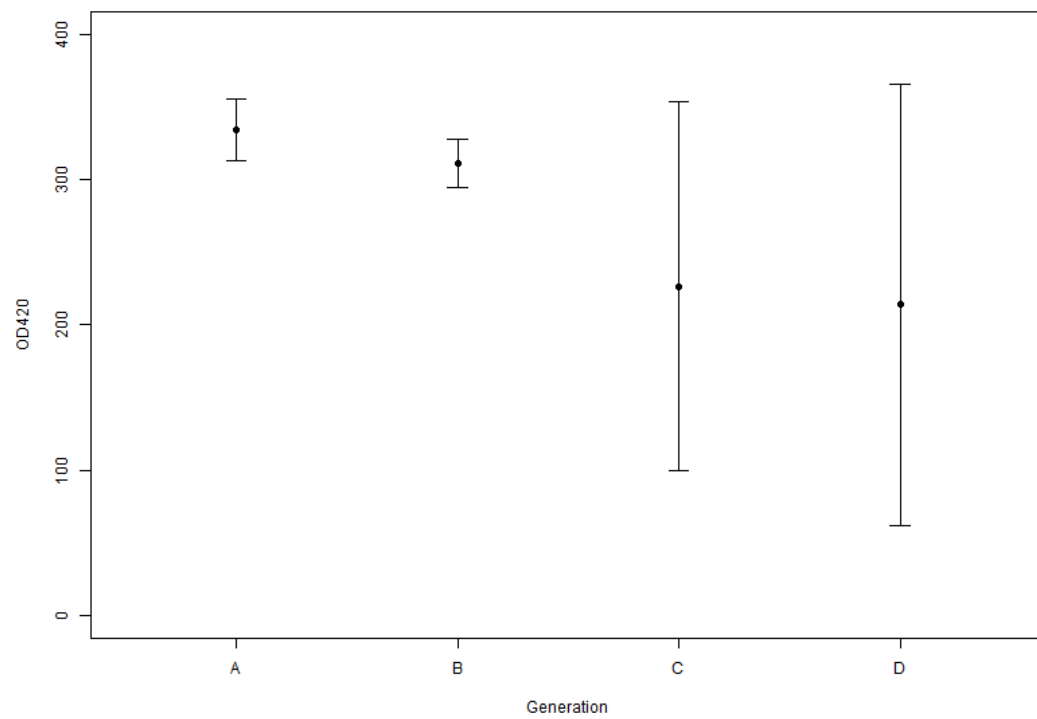
TABLE-1: mean and standard deviation for each generation in each replicant.

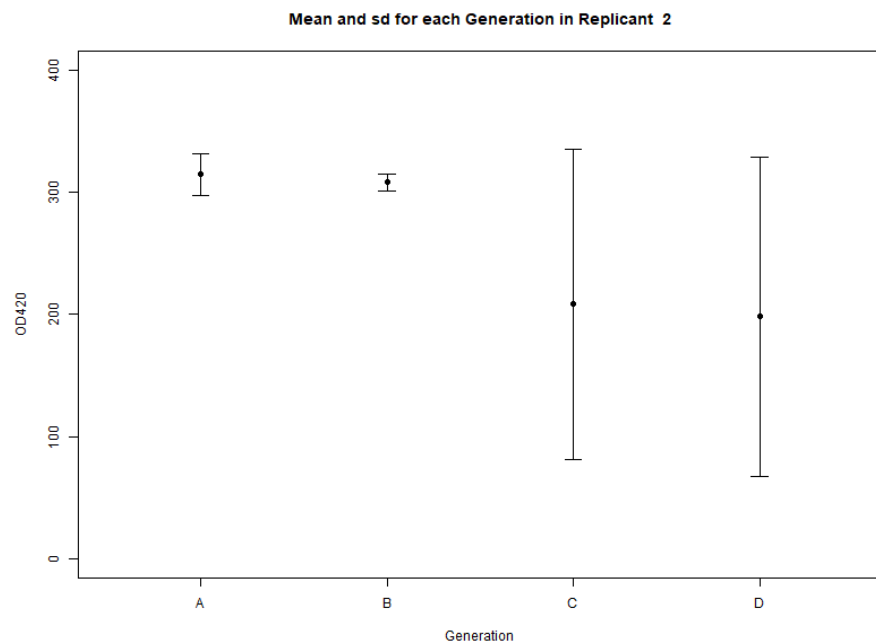
	Generation	Replicant	mean	stddev
1	A	1	334.7	21.3
2	A	2	314.5	17.1
3	A	3	319.0	9.9
4	A	4	302.0	6.2
5	A	5	330.2	5.0
6	B	1	310.9	16.5
7	B	2	308.0	7.1
8	B	3	304.8	8.3
9	B	4	288.5	7.5
10	B	5	319.3	13.1
11	C	1	226.3	126.9
12	C	2	208.2	127.1
13	C	3	194.8	125.9
14	C	4	186.8	122.2
15	C	5	209.4	137.0
16	D	1	213.8	152.1
17	D	2	198.0	130.8
18	D	3	187.5	139.0
19	D	4	172.6	131.1
20	D	5	195.8	149.2

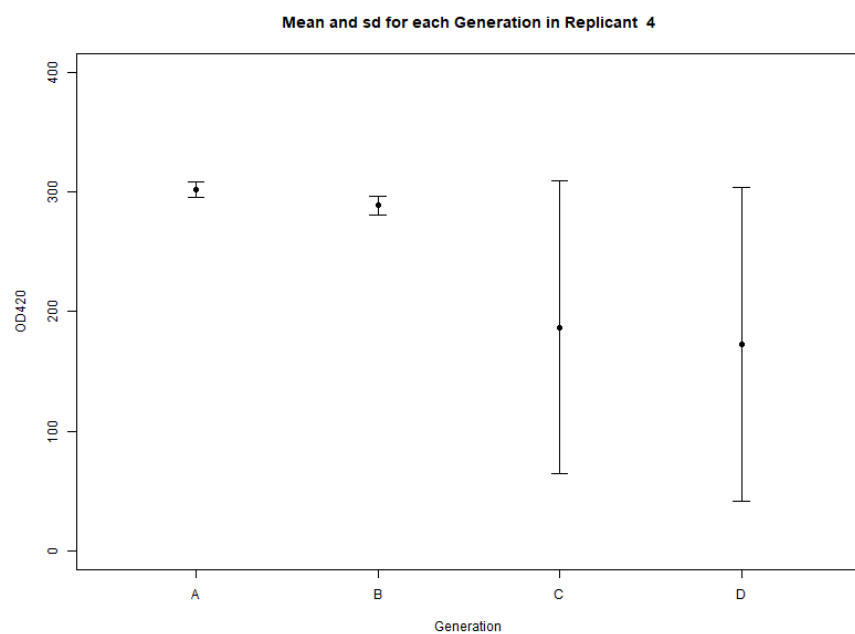
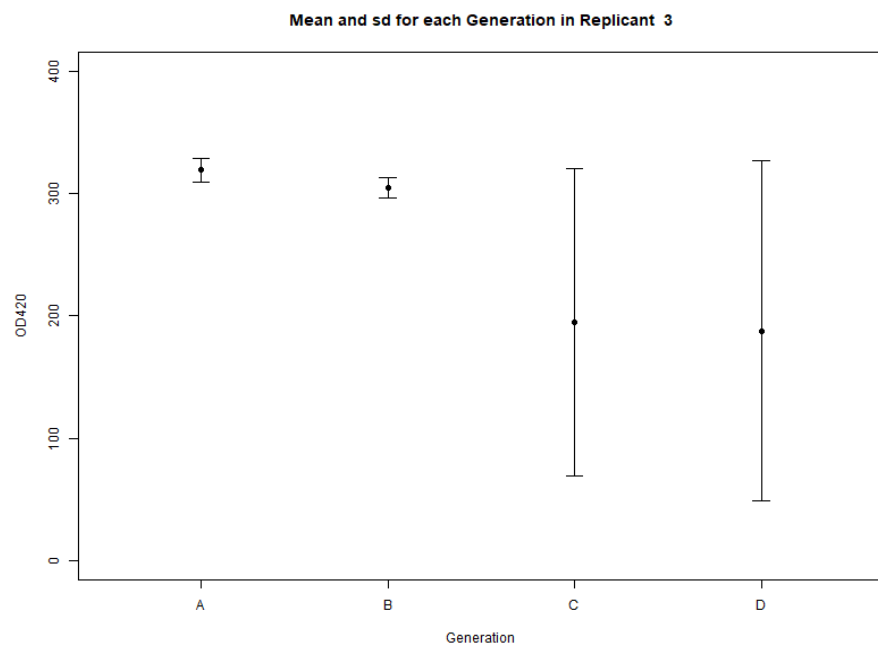
Mean and sd for each Generation in Replicant 5



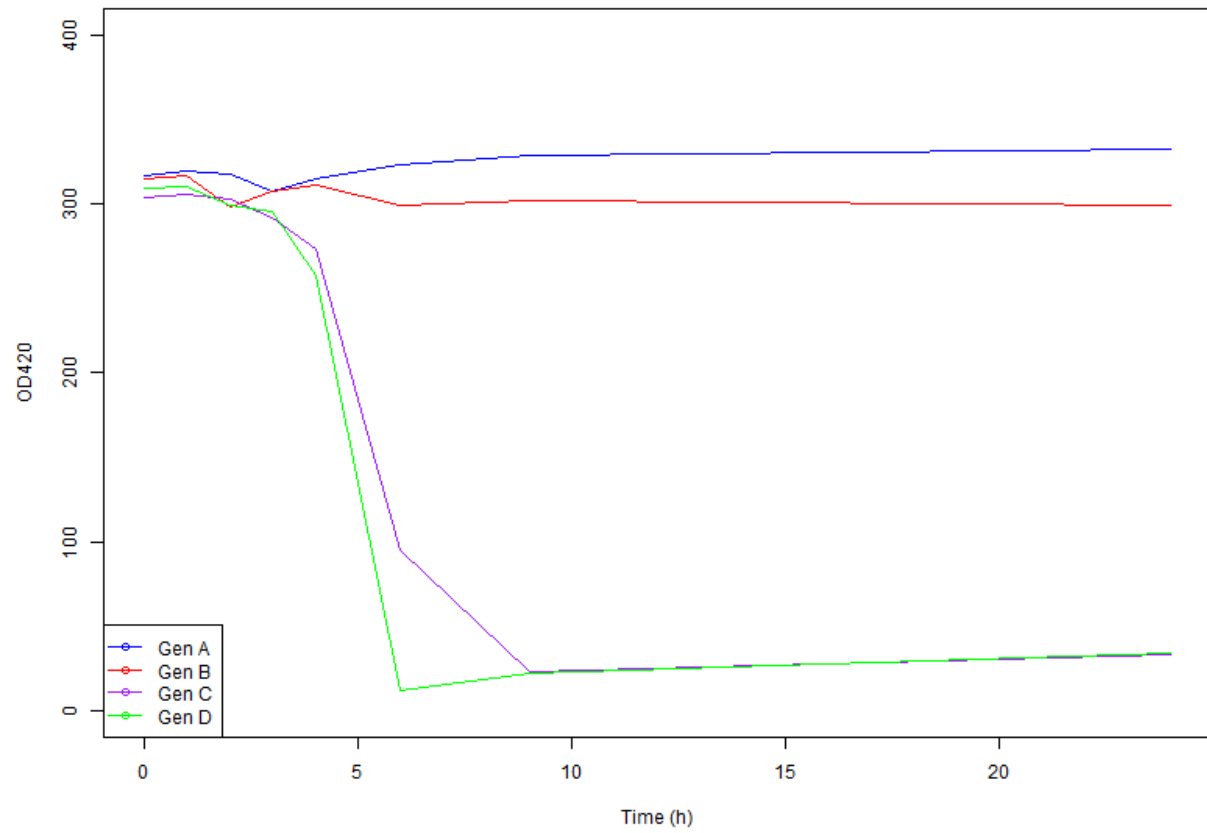
Mean and sd for each Generation in Replicant 1



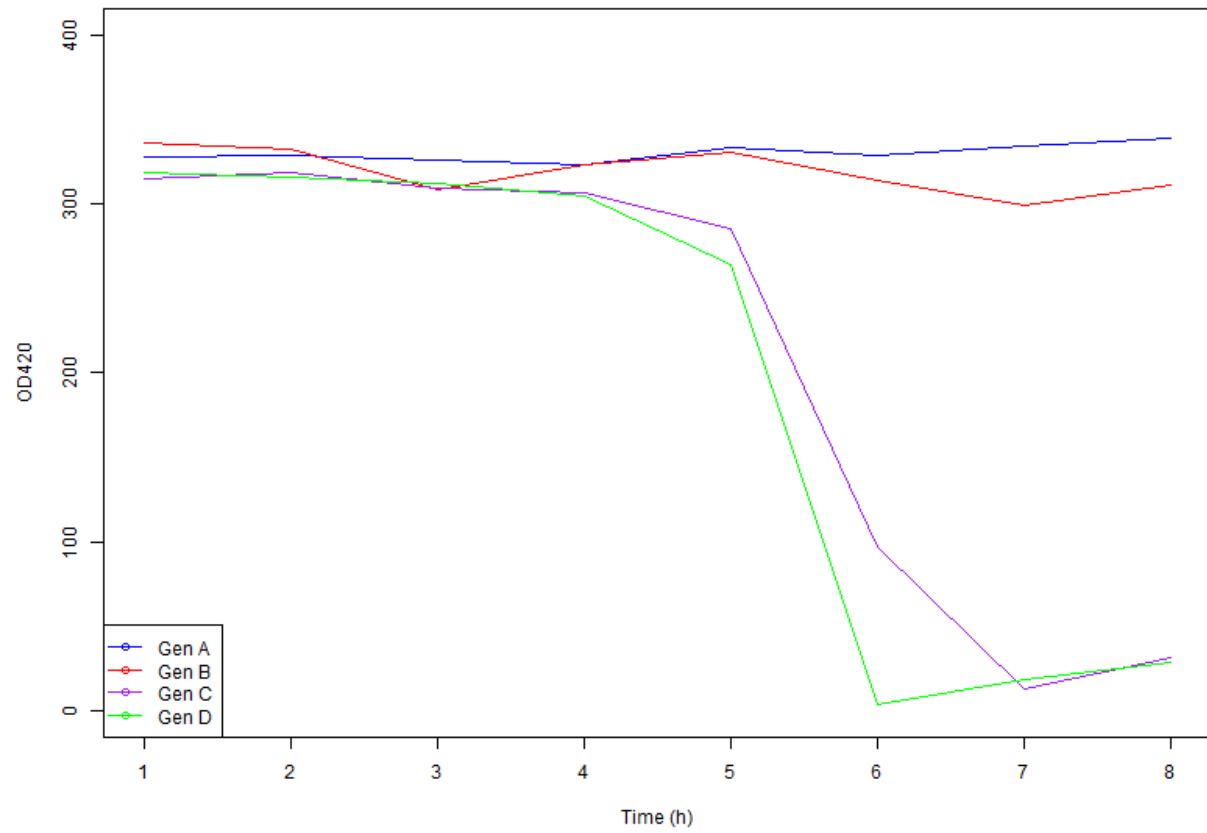




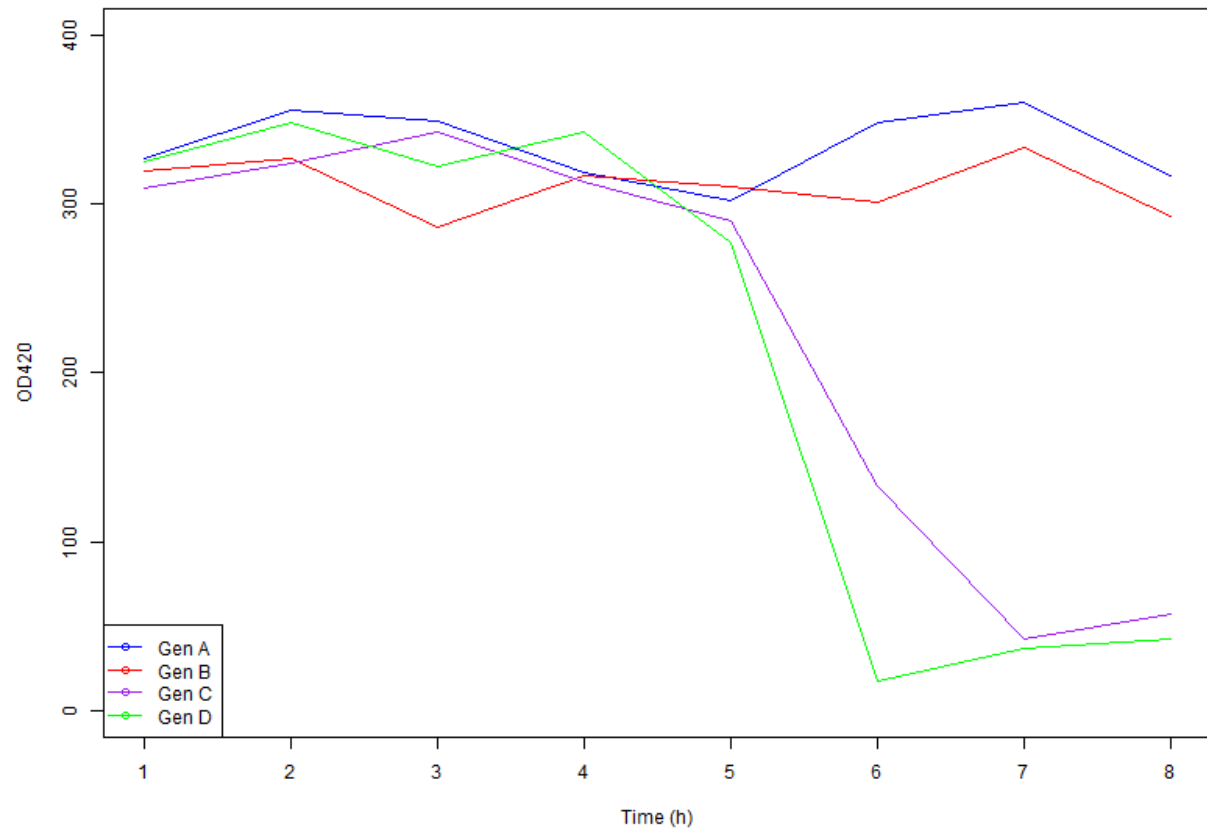
Plot-2: Citrate Concentrations in the culture medium of *E. coli*



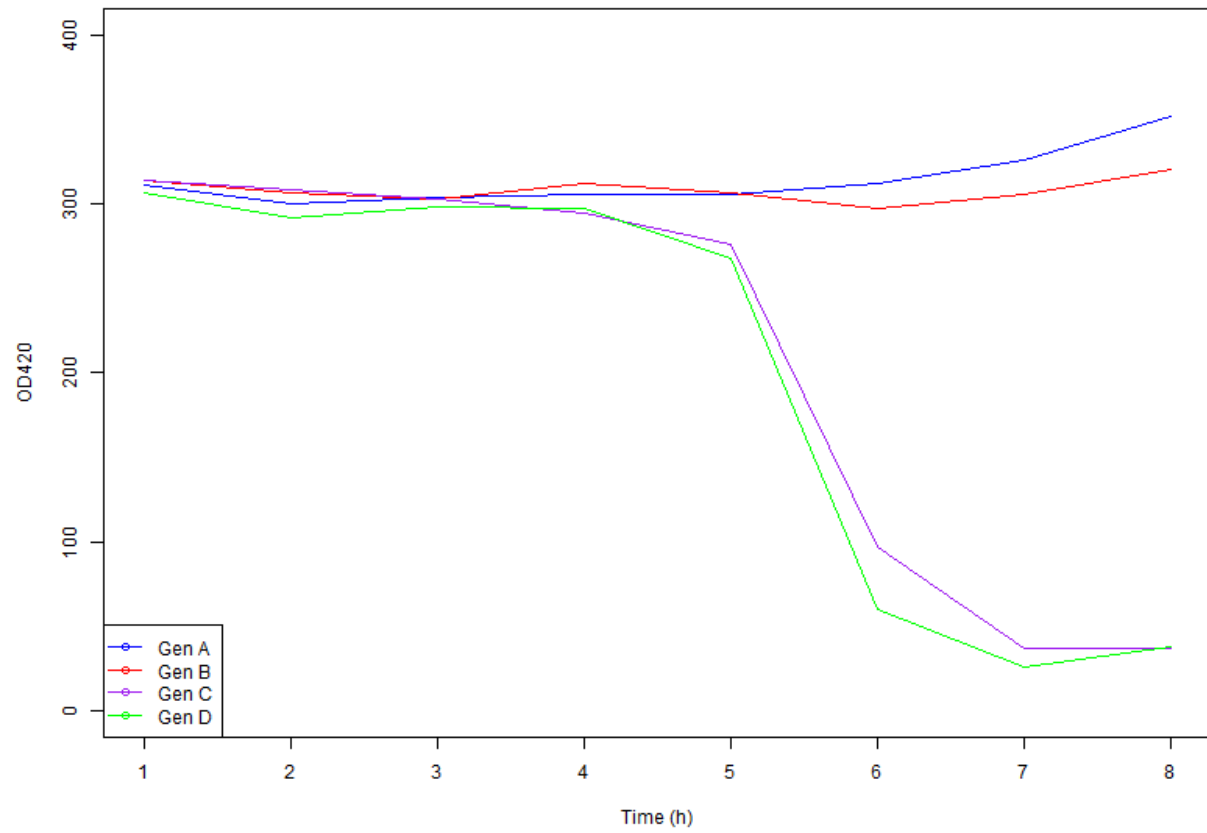
plot-1-5: Citrate Concentrations in the culture medium of *E. coli*



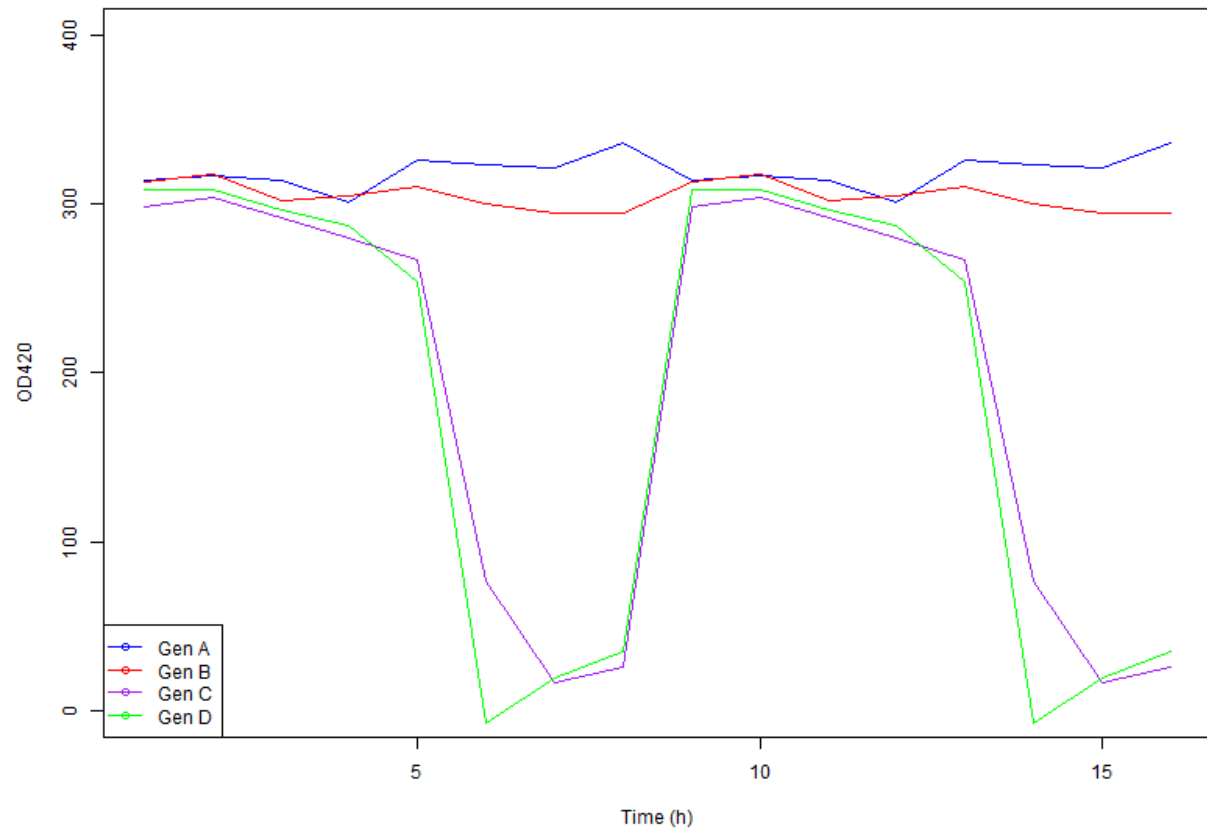
plot-1-1: Citrate Concentrations in the culture medium of *E. coli*



plot-1-2: Citrate Concentrations in the culture medium of *E. coli*



plot-1-3: Citrate Concentrations in the culture medium of *E. coli*



plot-1-4: Citrate Concentrations in the culture medium of *E. coli*

